

 **Day of Data Toronto 2026**

# Automating SQL Server at Scale with Ansible A

**Marcelo Adade**



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

# Our Sponsors



Toronto Data Professionals Community (TDPC)



# Community Support

[Toronto Data Professionals Community \(TDPC\)](#), one of the largest data professional's community in Toronto, host monthly event which offers interactive learning built by community and guided by trusted data experts.

TDPC Event Partners



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

# Sp\_help(DBA)

## Marcelo Adade

**Microsoft Data Platform MVP**

**Principal database Engineer (SQL Server)**

**Database Consultant and Instructor at DBBITS (Sao Paulo)**

**SQL Saturday São Paulo – organizer**

**<https://bit.ly/ansisqlyyz2026>**

**[marceloadade@gmail.com](mailto:marceloadade@gmail.com)**



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

## PART 1

# The problem

What goes wrong when there is no standard



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

# The script that ate the team

*It starts well. Somebody writes a PowerShell script that configures a new SQL Server correctly.*

1

**It gets copied to a second server**

Someone edits it to fit.

2

**It gets emailed to a colleague**

Someone edits it again.

3

**A new DBA writes their own**

Because they could not find yours.

4

**Three years pass**

**Nobody can tell you which version ran on which server.**



# What that looks like on Monday

You have twelve SQL Servers. The manager (possibly) asks a reasonable question:

*“Are we consistent?”*

To answer it you would have to log into twelve machines and read twelve sets of settings by hand.

**So nobody asks, and nobody answers.**

The settings quietly drift apart. That is called configuration drift, and it is the normal state of an estate nobody automated.



# Two servers. Five differences.

| Setting                        | sqlwin01  | sqlwin02 | Should be |
|--------------------------------|-----------|----------|-----------|
| Max degree of parallelism      | 0         | 8        | 8         |
| Cost threshold for parallelism | 5         | 50       | 50        |
| Max server memory              | unlimited | 24576 MB | 24576 MB  |
| Backup compression             | off       | on       | on        |
| Trace flag 3226                | off       | on       | on        |

**Nobody did this on purpose.**



# It gets worse, not better

1

## Drift is silent

Nothing alerts you. It shows up as a slow query at 2am.

2

## The fix does not stick

You correct a server today; the next rebuild brings the old settings back.

3

## It blocks everything else

You cannot patch confidently, or build an availability group, on servers you cannot describe.

**The problem is not that the scripts are bad. It is that nothing says what “correct” means.**



## PART 2

# A What Ansible is



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

# Ansible (from Red Hat)

1

## An inventory

A list of your servers.

`inventory.yml`

2

## Variables

The settings those servers should have.

`group_vars/`

3

## Playbooks

A list of tasks to run.

`playbooks/`

You run a playbook from your laptop. Ansible connects to each server, does what it needs to, and disconnects.



# Nothing is installed on the servers

## WINDOWS

### WinRM

Windows' own remote management service.

## LINUX

### SSH

The one you already use every day.

Both are already part of the operating system.

**No agent. No service. No scheduler.**

Nothing to deploy before you start, and nothing to uninstall if you stop.



# Idempotency

An idempotent task is one you can run repeatedly, safely. Run a playbook twice and the second run reports:

```
ok=7    changed=0
```

Nothing happened, because there was nothing left to do.

That is the whole argument against the copied script.

A script **does things**. A playbook **makes things true** — and doing nothing is a valid outcome.



# Ansible does not make your tools idempotent

It gives you somewhere to put the check.

**setup.exe is NOT idempotent. Run it twice by hand and it errors.**

The playbook checks first:

```
- name: Is SQL Server already installed?
  ansible.windows.win_reg_stat:
    path: HKLM:\SOFTWARE\Microsoft\Microsoft SQL Server\...
  register: installed
```

**A human wrote that guard. That is the job.**



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

# The build standard becomes a file

```
# group_vars/all.yml
baseline_maxdop: 8
baseline_cost_threshold: 50
baseline_max_memory_mb: 24576
baseline_backup_compression: true
baseline_trace_flag_3226: true
```

**This is the answer to  
“are we consistent?”**

Change a number here and you have  
changed the standard for the whole estate.

**It is reviewed in a pull request. It is not a script — it is **numbers**.**



## PART 3

# The demos

Six playbooks, read top to bottom



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

# Look, don't touch

Ask every server what version it is running, what its collation is, whether it is clustered.

**1****Safe to run anywhere**

Including production, on your first day.

**2****Needs no approvals**

It reads. It changes nothing.

**3****Immediate value**

You get an inventory nobody had before.



# Install SQL Server 2025

## WINDOWS

- Mount the ISO
- Generate the answer file from a template
- Run setup.exe
- Delete the answer file

## LINUX

- Add Microsoft's repository
- Install the mssql-server package
- Run mssql-conf setup
- Install sqlcmd and bcp



# Drift, then enforcement

```
ansible-playbook playbooks/3_baseline.yml --check --diff
```

--check changes nothing. It tells you what it would change.

**That is a drift report — and you did not have to build one.**

Then drop --check and run it again. Now the estate matches the file.

*Honest caveat: check-mode support varies by module, so a clean check is not a guarantee.*



# Availability group, in five layers

- 1 win\_feature — Failover Clustering
- 2 **The Windows cluster itself**
- 3 **The mirroring endpoint — hand-written T-SQL**
- 4 availability\_group and ag\_replica
- 5 Adding databases

**Nobody's collection builds you a Windows failover cluster 😊**



# Patching without logging in

Two lines carry this entire playbook:

```
serial: 1                # one server at a time, finish, then next
max_fail_percentage: 0    # one fails, stop. Do not touch the rest.
```

Without those, “run this everywhere at once” is how you take an entire estate down in ninety seconds.

## Blast-radius control

The difference between automation a DBA will allow and automation they will ban.



# Tools, with no internet

SSMS 22 installs through the Visual Studio Installer, which normally downloads as it goes. Locked-down servers do not allow that.

**1****Build an offline layout once**

A folder holding everything it would have downloaded.

**2****Copy it to the servers**

Then install with `--noWeb` so it never reaches out.

**The thing you install is not always one .exe.**



# Where to start

*Deliberately backwards from what you would expect.*

1

## Inventory the fleet with a read-only playbook

No approvals needed. Immediate value.

2

## Write the build standard as variables

Run it in check mode — that is a drift report.

3

## Enforce it on dev first

Let changed=0 convince the change board.

4

## Then automate builds and patching

**Installation is the flashy part. It is also the last part.**



# References

- <https://www.youtube.com/watch?v=QVNqYWbYwno>
- <https://github.com/automatesql/Ansible-for-SQL-Server-DBAs>
- <https://docs.ansible.com/projects/ansible/latest/collections/lowlydba/sqlserver/index.html>
- <https://galaxy.ansible.com/ui/repo/published/microsoft/sql/content/role/server/>
- <https://learn.microsoft.com/en-us/sql/linux/install-upgrade/deploy-ansible?view=sql-server-ver17>
- <https://docs.ansible.com/>



# Thank you

Marcelo Adade · @MARCELOADADE

<https://bit.ly/ansisqlyyz2026>

Every playbook is commented for somebody who has never used Ansible.  
Take it, change the numbers, delete what you do not need.

## *Questions?*



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)